

Current

The rate of flow of charge through any area of cross-section of a conductor is called current or electric current

Let, an infinitesimal amount of charge dq is passing through any area of a conductor in a very small interval of time dt , then by the definition of electric current in the conductor is

$$i = \frac{dq}{dt}$$

If the rate of flow of charge is constant, then the current is said to be steady (constant) current. The steady current in the conductor is denoted by I .

$$I = \frac{q}{t}$$

Where, q is the total charge that flows in time t .

The SI unit of current is *Ampere*.

$$1 \text{ Ampere} = \frac{1 \text{ Coulomb}}{1 \text{ second}}$$

Current density

The current density is the *electric current per unit area of cross-section*. The current density at a point is the *quantity of charge that flows through a unit area of cross-section that point per unit time*. It is denoted by J .

We can write the amount of current through an element as $\vec{J} \cdot d\vec{A}$ (Where $d\vec{A}$ is the vector area of the element). The total current through a surface is then

$$I = \int \vec{J} \cdot d\vec{A}$$

If the current is distributed uniformly across the surface and parallel to dA , then J is uniform and parallel to dA . Then

$$I = \int \vec{J} \cdot d\vec{A} = \int J dA = J \int dA = JA$$

$$\therefore J = \frac{I}{A}$$

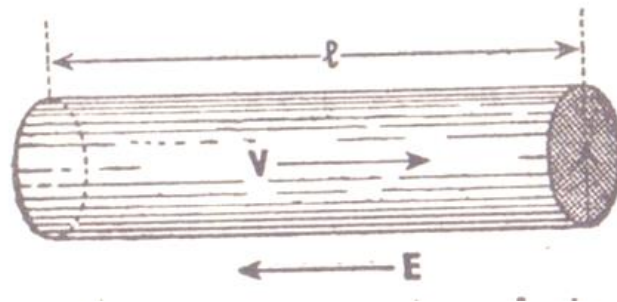
Where, A is the total surface area.

The SI unit of current density is *Ampere/m²*.

Drift velocity

The drift velocity is the average velocity of electron attains due to an electric field.

If the potential difference between two ends of a conductor is applied to the free electron, it is moved towards $-\vec{E}$, that is opposite to the field inside the conductor, with an average velocity v_d is called *drift velocity*.

Expression for drift velocity:**Fig. 1**

If N is the number of conduction electrons per unit volume of a conductor then the number of conduction electrons n inside the conductor of volume Al (Fig. 1.), is given by

$$n = NAl \quad (1)$$

Therefore, the total charge q is given by

$$q = NAle \quad (2)$$

Where, e is the electronic charge. If charge q flows through the conductor of volume Al in time t , then on the average the electrons cover a distance l in time t , so that

$$t = \frac{l}{v_d}$$

According to the definition of current

$$I = \frac{q}{t}$$

$$I = \frac{NAle}{t}$$

$$I = \frac{(NAle)v_d}{l}$$

$$I = NAev_d$$

$$v_d = \frac{I}{NAe} \quad (3)$$

We know the current density

$$J = \frac{I}{A} \quad (4)$$

Comparing equations (3) and (4) we can write

$$v_d = \frac{J}{Ne} \quad (5)$$

Equation (5) is the required expression for the drift velocity of electrons in a conductor conducting N electrons per unit volume.

Resistance

Resistance is the property of a substance due to which it opposes the flow of electricity through it. It is denoted by R .

It is measured by Ohm's law. According to Ohm's law

$$R = \frac{V}{I}$$

Where, I is the current and V is the potential difference.

The unit of resistance is Ohm (Ω).

Resistivity

Resistivity is the property of the material of a conductor. The *electric resistance of a conductor is directly proportional to the length of the conductor and inversely proportional to the cross-sectional area of the conductor*. This relation is expressed as

$$R \propto \frac{l}{A}$$

Where, l is the length of the conductor and A is the cross-sectional area of the conductor. We can write,

$$R = \text{Constant} \times \frac{l}{A}$$

$$R = \rho \frac{l}{A}$$

Where, ρ is called the *resistivity* of the material and is constant for given material.

If $l = 1 \text{ m}$ and $A = 1 \text{ m}^2$ then $R = \rho$. Thus the resistivity is defined as the resistance of a material of unit cross-section and unit length.

The unit of resistivity is Ohm-meter ($\Omega\text{-m}$) or Ohm-centimeter ($\Omega\text{-cm}$).

Conductance

Conductance is the reciprocal of resistance where the resistance of a conductor measures the opposition which it offers to the flow of current, the conductance measures the inducement which it offers to its flow.

If the conductance of a current be represented by G and its resistance by R , then

$$G = \frac{1}{R}$$

$$G = \frac{A}{\rho l} = \frac{1}{\rho} \frac{A}{l}$$

$$G = \sigma \frac{A}{l}$$

Where, σ is the conductivity and

$$\sigma = \frac{1}{\rho}$$

Thus the conductance is directly proportional to the cross-sectional area of the conductor and inversely proportional to length of the conductor.

Conductivity

Conductivity is the property of a conductor by means of which it allowed the current to flow through the conductor when the potential difference is applied across the conductor. It is the reciprocal of resistivity and is represented by σ .

$$\sigma = \frac{1}{\rho}$$

The unit of conductivity is $\text{Ohm}^{-1}\text{-meter}^{-1}$ or mho/meter.

Ohm's law

Statement: The potential difference between the ends of a conductor varies directly as the current flowing in it, provided the temperature does not change and the physical state of the conductor remains the same.

Explanation: If the current flowing through the conductor is I and the potential difference between the ends of the conductor is $V_A - V_B$, then we can write

$$(V_A - V_B) \propto I$$

$$(V_A - V_B) = \text{Constant} \times I$$

$$\therefore \frac{(V_A - V_B)}{I} = \text{Constant}$$

The ratio is *constant* and is known as the *resistance* of the material.

Therefore we can write

$$\frac{(V_A - V_B)}{I} = R$$

Denoting $V = V_A - V_B$, we can write

$$\frac{V}{I} = R$$

$$V = IR$$

This is known as Ohm's law.

From the above equation we can write

$$\frac{V}{I} = \rho \frac{l}{A}$$

$$\frac{V}{l} = \rho \frac{I}{A}$$

$$E = \rho J$$

This is another form of Ohm's law.

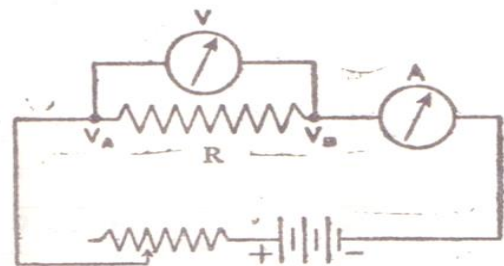


Fig. 2

Resistivity-an atomic view

The conduction electrons in the metal are *free to move*. We also assume that the electrons collide not with *one another* but only with *positive charge* of the metal.

If an electron of mass m is placed in an electric field of magnitude E , the electron will experience an acceleration given by Newton's second law

$$a = \frac{F}{m} = \frac{eE}{m}$$

In the average time τ between collisions, the average electron will acquire a *drift speed* of $v_d = a\tau$. Thus, at any instant, on average, the electrons will have drift speed

$$v_d = a\tau = \frac{eE\tau}{m}$$

We know,

$$v_d = \frac{J}{Ne}$$

Comparing above two equations

$$\frac{J}{Ne} = \frac{eE\tau}{m}$$

$$\therefore E = \left(\frac{m}{Ne^2\tau} \right) J$$

We know

$$E = \rho J$$

Comparing above two equations

$$\rho = \frac{m}{Ne^2\tau}$$

In the above equation N , the number of conduction electrons per volume, is *independent of the field*, and m and e are *constants*. The average time between collisions τ , is a constant, because the drift speed v_d is so much smaller than the effective speed v_{eff} of the electrons. Thus, for metals their resistivity ρ is a constant.

Thus we can say that *metals obey Ohm's law*.

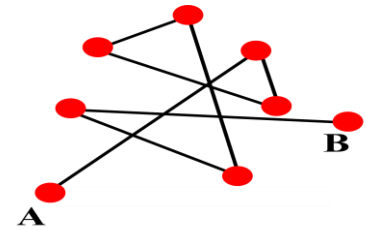


Fig. 3. An electron moving from A to B, making six collisions.

Mathematical Problems

Problem-1: What is the drift velocity of electrons in a copper conductor having a cross-sectional area of $5 \times 10^{-6} \text{ m}^2$, if the current is 10 A? Assume that there are $8 \times 10^{28} \text{ electrons/m}^3$ ($q_e = 1.6 \times 10^{-19} \text{ C}$).

Solution: We know

$$v_d = \frac{I}{NAe}$$

$$\text{Or } v_d = \frac{10}{8 \times 10^{28} \times 5 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

$$\therefore v_d = 1.56 \times 10^{-4} \text{ ms}^{-1} \text{ (Ans)}$$

Here,

$$A = 5 \times 10^{-6} \text{ m}^2$$

$$I = 10 \text{ Amp}$$

$$N = 8 \times 10^{28} \text{ electrons/m}^3$$

$$e = q_e = 1.6 \times 10^{-19} \text{ C}$$

$$v_d = ?$$

Problem-2: A silver wire 1 mm in diameter carries a charge of 90 coulomb in 1 hr. Silver contains $5.8 \times 10^{28} \text{ free electrons/m}^3$. Calculate (i) the current in the wire (ii) current density and drift velocity of electrons in the wire.

Solution: We know, Current

$$I = \frac{q}{t} = \frac{90}{3600} = 0.025 \text{ amp (Ans)}$$

Current density

$$J = \frac{I}{A} = \frac{0.025}{7.85 \times 10^{-7}}$$

$$= 3.18 \times 10^{-4} \text{ amp/m}^2 \text{ (Ans)}$$

Drift velocity

$$v_d = \frac{I}{NAe}$$

$$\text{Or } v_d = \frac{0.025}{5.8 \times 10^{28} \times 7.85 \times 10^{-7} \times 1.6 \times 10^{-19}}$$

$$\therefore v_d = 3.43 \times 10^{-6} \text{ ms}^{-1} \text{ (Ans)}$$

Here,

$$A = \pi r^2 = 3.1416 \times \left(\frac{1 \times 10^{-3}}{2} \right)^2$$

$$= 7.85 \times 10^{-7} \text{ m}^2$$

$$q = 90 \text{ Coulomb}$$

$$t = 1 \text{ hr} = 3600 \text{ s}$$

$$N = 5.8 \times 10^{28} \text{ electrons/m}^3$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$I = ?$$

$$v_d = ?$$